CHEMISTRY 101 · ANSWER KEY - PROFESSOR COPY

Homework 1: Atoms & Air

Grading guide. Total **23 points**. Answers in different words that mean the same thing get full credit; partial credit is suggested where it applies. Enter the final score in Professor mode on the recitation page.

Part A — Count the atoms

1. For each molecule, count the atoms of each element, then the total. (Remember: the little number counts the atom right before it; no number means 1. Write 0 if an element isn't there.) **(6 pts)**

Answer: H_2O: C0 H2 O1 N0 = 3. CO_2: C1 H0 O2 N0 = 3. O_2: 0,0,2,0 = 2. N_2: 0,0,0,2 = 2. NH_3: 0,3,0,1 = 4. $C_6H_{12}O_6$: 6,12,6,0 = 24. One point per correct row.

Part B — Draw a molecule

2. Draw CO_2 and H_2O as ball diagrams, like the ones in class. Use the legend: oxygen = red (or label "O"), carbon = black ("C"), hydrogen = light blue ("H"). Label each molecule with its formula. **(2 pts)**

Answer: CO_2: one C ball with an O ball on each side (O-C-O). H_2O: one O ball with two smaller H balls attached (a V shape is a bonus, not required). 1 point each; the correct number and kind of balls is what matters.

Part C — The air, in percent

3. Out of every 100 particles of air, about how many are nitrogen? About how many are oxygen? **(2 pts)**

Answer: About 78 nitrogen; about 21 oxygen.

4. Imagine a room with 100 balloons of air. If 21 of them are oxygen, how many balloons are NOT oxygen? **(1 pt)**

Answer: 79 (100 - 21).

5. A NICU baby is "on 40%." Is that MORE or LESS oxygen than room air? How do you know? **(2 pts)**

Answer: More. Room air is 21% oxygen; 40% means the blender has added extra oxygen on top of that.

Part D — Which way does it go?

6. In the lungs, mark what happens to each gas with an X: goes INTO the blood, goes OUT of the blood into the air, or is mostly ignored (in and right back out). (3 pts)

Answer: O_2 -> INTO the blood. CO_2 -> OUT of the blood. N_2 -> mostly ignored.

7. Why must the body get rid of carbon dioxide? (1 pt)

Answer: A. Dissolved in blood it forms an acid, and too much acid disrupts the body's chemistry. - CO_2 in blood makes acid. A little is normal; a pile-up (when breathing fails) makes blood too acidic.

Part E — Flashback to NICU Tutor

8. Name the NICU machine that measures oxygen saturation using light. (1 pt)

Answer: The pulse oximeter.

9. What is surfactant, and why do premature babies need it as a medicine? (2 pts)

Answer: A slippery, soap-like liquid that coats the alveoli so they don't stick shut. Babies don't make enough until about 34–36 weeks, so preemies' lungs collapse with each breath (RDS) unless surfactant is given.

Part F — Think like a doctor

10. A preemie's blood gas test shows too much carbon dioxide in the blood. (a) What is this baby probably NOT doing well enough? (b) Name one NICU machine that could help, and say in one sentence what it does. (3 pts)

Answer: (a) Breathing — not moving enough air in and out to blow off CO_2 (1 pt). (b) CPAP (keeps lungs partly inflated so breathing is easier) or a ventilator (delivers breaths for the baby) — 1 pt for the machine, 1 pt for what it does.